

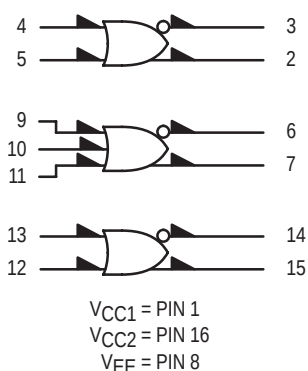
MC10105

Triple 2-3-2-Input OR/NOR Gate

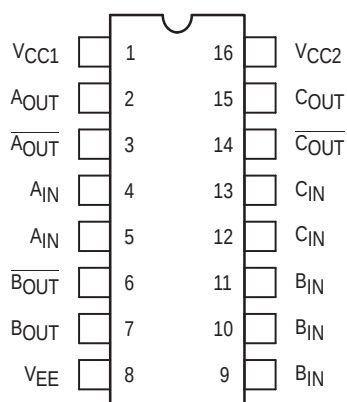
The MC10105 is a triple 2–3–2 input OR/NOR gate.

- $P_D = 30 \text{ mW typ/gate (No Load)}$
- $t_{pd} = 2.0 \text{ ns typ}$
- $t_r, t_f = 2.0 \text{ ns typ (20\%–80\%)}$

LOGIC DIAGRAM



DIP PIN ASSIGNMENT



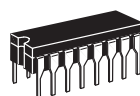
Pin assignment is for Dual-in-Line Package.
For PLCC pin assignment, see the Pin Conversion Tables on page 18.



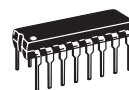
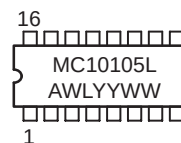
ON Semiconductor

<http://onsemi.com>

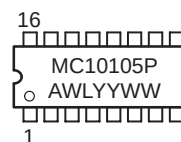
MARKING DIAGRAMS



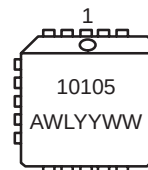
CDIP-16
L SUFFIX
CASE 620



PDIP-16
P SUFFIX
CASE 648



PLCC-20
FN SUFFIX
CASE 775



A = Assembly Location
 WL = Wafer Lot
 YY = Year
 WW = Work Week

ORDERING INFORMATION

Device	Package	Shipping
MC10105L	CDIP-16	25 Units / Rail
MC10105P	PDIP-16	25 Units / Rail
MC10105FN	PLCC-20	46 Units / Rail

ELECTRICAL CHARACTERISTICS

Characteristic	Symbol	Pin Under Test	Test Limits							Unit
			−30°C		+25°C			+85°C		
			Min	Max	Min	Typ	Max	Min	Max	
Power Supply Drain Current	I _E	8		23		17	21		23	mAdc
Input Current	I _{inH}	4		425			265		265	μAdc
	I _{inL}	4	0.5		0.5			0.3		μAdc
Output Voltage Logic 1	V _{OH}	3 2	−1.060 −1.060	−0.890 −0.890	−0.960 −0.960		−0.810 −0.810	−0.890 −0.890	−0.700 −0.700	Vdc
Output Voltage Logic 0	V _{OL}	3 2	−1.890 −1.890	−1.675 −1.675	−1.850 −1.850		−1.650 −1.650	−1.825 −1.825	−1.615 −1.615	Vdc
Threshold Voltage Logic 1	V _{OHA}	3 2	−1.080 −1.080		−0.980 −0.980			−0.910 −0.910		Vdc
Threshold Voltage Logic 0	V _{OLA}	3 2		−1.655 −1.655			−1.630 −1.630		−1.595 −1.595	Vdc
Switching Times (50Ω Load)										ns
Propagation Delay	t _{4+3−}	3	1.0	3.1	1.0	2.0	2.9	1.0	3.3	
	t _{4−3+}	3	1.0	3.1	1.0	2.0	2.9	1.0	3.3	
	t ₄₊₂₊	2	1.0	3.1	1.0	2.0	2.9	1.0	3.3	
	t _{4−2−}	2	1.0	3.1	1.0	2.0	2.9	1.0	3.3	
Rise Time (20 to 80%)	t ₃₊	3	1.1	3.6	1.1	2.0	3.3	1.1	3.7	
	t ₂₊	2	1.1	3.6	1.1	2.0	3.3	1.1	3.7	
Fall Time (20 to 80%)	t _{3−}	3	1.1	3.6	1.1	2.0	3.3	1.1	3.7	
	t _{2−}	2	1.1	3.6	1.1	2.0	3.3	1.1	3.7	

MC10105

ELECTRICAL CHARACTERISTICS (continued)

@ Test Temperature			TEST VOLTAGE VALUES (Volts)					(V _{CC}) Gnd
			V _{IHmax}	V _{ILmin}	V _{IHAmin}	V _{ILAmax}	V _{EE}	
			–30°C	–0.890	–1.890	–1.205	–1.500	–5.2
			+25°C	–0.810	–1.850	–1.105	–1.475	–5.2
			+85°C	–0.700	–1.825	–1.035	–1.440	–5.2
Characteristic	Symbol	Pin Under Test	TEST VOLTAGE APPLIED TO PINS LISTED BELOW					(V _{CC}) Gnd
			V _{IHmax}	V _{ILmin}	V _{IHAmin}	V _{ILAmax}	V _{EE}	
Power Supply Drain Current	I _E	8					8	1, 16
Input Current	I _{inH}	4	4				8	1, 16
	I _{inL}	4		4			8	1, 16
Output Voltage Logic 1	V _{OH}	3					8	1, 16
		2	4				8	1, 16
Output Voltage Logic 0	V _{OL}	3	4				8	1, 16
		2					8	1, 16
Threshold Voltage Logic 1	V _{OHA}	3				4	8	1, 16
		2			4		8	1, 16
Threshold Voltage Logic 0	V _{OLA}	3			4		8	1, 16
		2				4	8	1, 16
Switching Times (50Ω Load)					Pulse In	Pulse Out	–3.2 V	+2.0 V
Propagation Delay	t _{4+3–}	3			4	3	8	1, 16
	t _{4–3+}	3			4	3	8	1, 16
	t ₄₊₂₊	2			4	2	8	1, 16
	t _{4–2–}	2			4	2	8	1, 16
Rise Time (20 to 80%)	t ₃₊	3			4	3	8	1, 16
	t ₂₊	2			4	2	8	1, 16
Fall Time (20 to 80%)	t _{3–}	3			4	3	8	1, 16
	t _{2–}	2			4	2	8	1, 16

Each MECL 10,000 series circuit has been designed to meet the dc specifications shown in the test table, after thermal equilibrium has been established. The circuit is in a test socket or mounted on a printed circuit board and transverse air flow greater than 500 linear fpm is maintained. Outputs are terminated through a 50-ohm resistor to –2.0 volts. Test procedures are shown for only one gate. The other gates are tested in the same manner.